Assignment – 3

MCA

Machine Learning (MCA201T)

Question 1 (5 marks)

Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.

Answer:

Supervised learning is a type of machine learning in which the model is trained on a **labelled dataset**, i.e. every training example consists of input features (X) and the correct output or label (y). The algorithm learns a mapping function y = f(X) from these examples, and after training it uses this function to predict the output for new, unseen data. It is called "supervised" because the known labels act like a teacher: the model compares its prediction with the correct answer, calculates the error and keeps adjusting itself to reduce that error.

Unsupervised learning, on the other hand, works on **unlabelled data**. No correct answers are given, so the algorithm itself has to explore the data and find hidden patterns, groups or structure.

Key differences

Basis	Supervised Learning	Unsupervised Learning
Type of data	Labelled data (X and y)	Unlabelled data (only X)
Main goal	Predict the output for new inputs	Discover hidden patterns or groups
Feedback	Direct feedback by comparing with true labels	No feedback / no correct answer
Tasks	Classification and Regression	Clustering, association, dimensionality reduction
Evaluation	Easy, using metrics like accuracy or MSE	Difficult, as there is no ground truth
Algorithms	Linear Regression, Decision Tree, SVM	K-Means, Hierarchical Clustering, PCA
Example	Predicting loan default from past customers whose result is known	Grouping shopping-app customers into segments using K-Means

Example of supervised learning: A bank has records of past customers along with a label showing whether each customer repaid the loan or defaulted. A model trained on this data can predict whether a new applicant will default. **Example of unsupervised learning:** An online shopping company has purchase data of customers but no predefined groups. K-Means clustering can divide the customers into groups such as frequent buyers, discount seekers and occasional buyers.

Question 2 (5 marks)

What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.

Answer:

Supervised learning problems are mainly of two types: **Classification** and **Regression**. The difference depends on the kind of output the model has to predict.

1. Classification: The output variable is a **discrete category or class**. The model learns from labelled examples and decides which class a new input belongs to. If there are only two classes (yes/no, spam/not spam) it is called binary classification; if there are more than two classes (digits 0–9) it is multi-class classification. Common algorithms are Logistic Regression, Decision Tree, SVM, k-NN and Naive Bayes.

2. Regression: The output variable is a **continuous numerical value**. The model learns the relationship between the input features and a quantity and predicts that quantity for new inputs. Common algorithms are Linear Regression, Polynomial Regression, Ridge/Lasso and Random Forest Regressor.

Basis	Classification	Regression
Output	Discrete class label	Continuous numeric value
Aim	Assign input to one of the predefined classes	Predict a quantity
Decision boundary / line	Separates the classes	Fits a line/curve through the data
Evaluation metrics	Accuracy, Precision, Recall, F1, ROC-AUC	MSE, RMSE, MAE, R^2
Real-world example	Email spam detection (spam / not spam)	Predicting price of a house from area, location and rooms

Question 3 (5 marks)

Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define the terms (dependent variable, independent variable, intercept, and slope). What is the role of the residual/error term?

Answer:

Linear Regression is a supervised learning algorithm used to predict a continuous value. It assumes that there is a linear (straight-line) relationship between the input variable(s) and the output variable, and it finds the straight line that fits the data best. The best-fit line is obtained by the **method of least squares**, which chooses the line for which the sum of squared differences between the actual and predicted values is minimum. When there is only one input variable it is called **simple linear regression**, and with more than one input it is called multiple linear regression.

Equation of simple linear regression

$$y = \beta 0 + \beta 1x + \varepsilon$$

- **Dependent variable (y):** the output or response that we want to predict, e.g. marks obtained by a student.
- **Independent variable (x):** the input or predictor used to explain/predict y, e.g. number of hours studied.
- **Intercept (β0):** the value of y when x = 0; it is the point where the line cuts the y-axis.
- **Slope (β1):** the change in y for every one-unit increase in x. A positive slope means y increases with x, a negative slope means y decreases as x increases.
- **Error term (ε):** the difference between the actual value and the value given by the line.

The values of the coefficients are calculated using:

$$\beta 1 = \Sigma(xi - \bar{x})(yi - \bar{y}) / \Sigma(xi - \bar{x})^2 \qquad \beta 0 = \bar{y} - \beta 1 \cdot \bar{x}$$

Role of the residual / error term

A residual is the difference between the actual value and the predicted value: **ei = yi − ŷi**. In real data the points never lie exactly on the line because of noise, measurement errors and factors that are not included in the model; the error term represents all this unexplained variation. Its roles are:

- The model is trained by minimising the **Sum of Squared Errors (SSE) = Σei²**.
- It measures how well the model fits the data: small residuals mean a good fit.
- Residual plots are used to check the assumptions of linear regression: residuals should be randomly scattered around zero, have constant variance and be approximately normally distributed.

Question 4 (5 marks)

How does Logistic Regression differ from Linear Regression? Write the logistic (sigmoid) function and explain why it is suitable for binary classification. What is the decision boundary in logistic regression?

Answer:

Linear regression predicts a continuous value, whereas **logistic regression** is a classification algorithm that predicts the **probability** that an input belongs to a particular class. Despite its name, logistic regression is used for classification, not for predicting numbers.

Basis	Linear Regression	Logistic Regression
Problem type	Regression	Classification (mostly binary)
Output	Any real value (−∞ to +∞)	Probability between 0 and 1
Equation	y = β0 + β1x	p = 1 / (1 + e^−(β0 + β1x))
Loss function	Mean Squared Error	Log loss (binary cross-entropy)
Parameter estimation	Least squares	Maximum likelihood

Basis	Linear Regression	Logistic Regression
Curve	Straight line	S-shaped curve

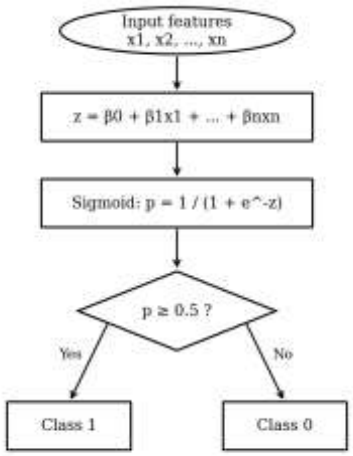


Fig. 1: Flowchart of Logistic Regression

Logistic (sigmoid) function

$$\sigma(z) = 1 / (1 + e^{(-z)}), \quad \text{where } z = \beta 0 + \beta 1x1 + \beta 2x2 + \ldots + \beta nxn$$

Why it is suitable for binary classification: The sigmoid function converts any real number z into a value between 0 and 1, so the output can be interpreted as the probability of class 1, i.e. P(y = 1 | x). For large positive z the output is close to 1 and for large negative z it is close to 0; at z = 0 it is exactly 0.5. The function is smooth and differentiable, so gradient descent can be used to train it. A straight line, in contrast, can give values below 0 or above 1, which cannot be probabilities.

Decision boundary

The decision boundary is the line (or surface) that separates the region where the model predicts class 1 from the region where it predicts class 0. With the usual threshold of 0.5, the model predicts class 1 when $\sigma(z) \geq 0.5$, and since $\sigma(z) = 0.5$ exactly when $z = 0$, the boundary is given by:

$$\beta 0 + \beta 1x1 + \beta 2x2 + \ldots + \beta nxn = 0$$

This is a straight line for two features, a plane for three features and a hyperplane for more features, so logistic regression is a linear classifier. If the threshold is changed (for example to 0.3), the boundary shifts, which changes the balance between precision and recall.

Question 5 (5 marks)

Describe how a Decision Tree works for classification. Explain the concepts of Information Gain and Gini Impurity. Why can decision trees overfit the training data, and how can this problem be reduced?

Answer:

Working of a decision tree

A decision tree is a flowchart-like model that classifies data by asking a series of questions about the features. It starts with the **root node**, which contains the complete training data. At every **internal node** the algorithm selects the feature and the condition that separates the classes best, and the data is divided into **branches**. This splitting is repeated recursively for each child node until the node becomes pure (contains only one class) or a stopping condition such as maximum depth is reached. The final nodes are called **leaf nodes**, and each leaf gives the predicted class (the majority class of the samples in it). To classify a new record, we start at the root and follow the branches according to the feature values until we reach a leaf.

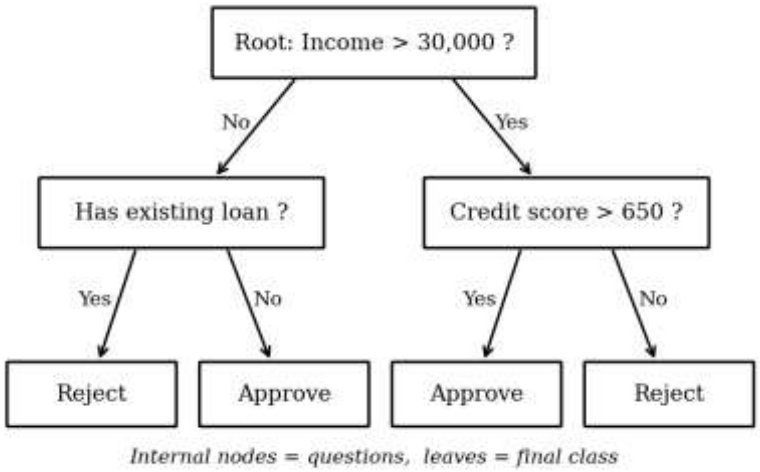


Fig. 2: Example of a Decision Tree (loan approval)

Information Gain

Information gain is based on **entropy**, which measures the impurity or randomness in a node. Entropy is 0 for a pure node and 1 for a binary node with a 50–50 split. Information gain is the reduction in entropy after splitting on an attribute A. The attribute with the **highest** information gain is chosen for the split (used in ID3 and C4.5).

$$Entropy(S) = -\Sigma\, pi \cdot log2(pi) \qquad IG(S, A) = Entropy(S) - \Sigma\, (|Sv| / |S|) \cdot Entropy(Sv)$$

Gini Impurity

Gini impurity measures the probability that a randomly chosen sample from the node would be wrongly classified if it were labelled according to the class distribution of the node. It is 0 for a pure node and 0.5 for a perfectly mixed binary node. The split with the **lowest** weighted Gini impurity is chosen (used in CART). It is computationally faster than entropy because no logarithm is needed.

$$Gini(S) = 1 - \Sigma\, pi^2$$

Overfitting and how to reduce it

If a tree is allowed to grow fully, it keeps splitting until every leaf is pure. In this process it memorises the noise and outliers of the training data, so it gives very high training accuracy but poor accuracy on new data (high variance). This problem can be reduced by:

- **Pre-pruning:** stopping the growth early by limiting max_depth, min_samples_split, min_samples_leaf or max_leaf_nodes.
- **Post-pruning:** growing the full tree and then removing branches that do not improve validation accuracy (e.g. cost-complexity pruning).
- Using **cross-validation** to select the best hyper-parameters, and using more training data.
- Using **ensemble methods** like Random Forest, which average many trees.

Question 6 (5 marks)

What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples of algorithms that use each technique. Why do ensemble methods generally perform better than a single model?

Answer:

Ensemble methods are techniques in which several individual models (called base learners or weak learners) are trained and their predictions are combined to get one final prediction. The combination is done by majority voting for classification or by averaging for regression. The idea is similar to taking the opinion of a group of experts instead of depending on one person.

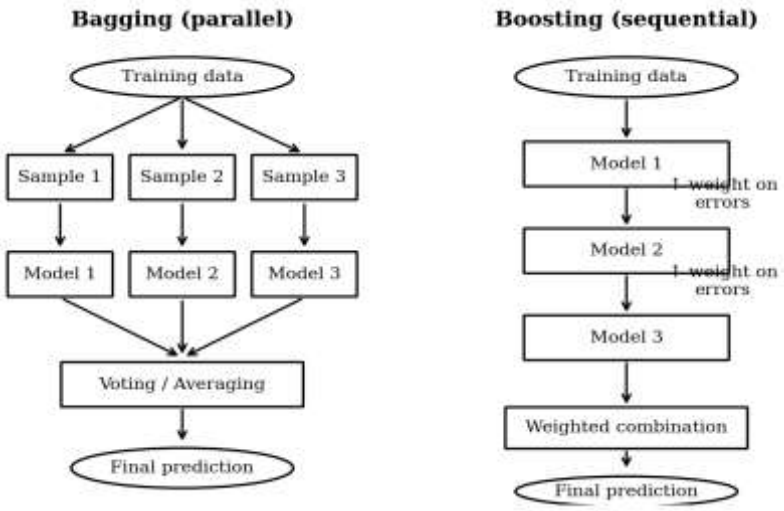


Fig. 3: Bagging vs Boosting

Basis	Bagging	Boosting
Full form / idea	Bootstrap Aggregating: train many models on random samples	Train models one after another, each correcting the previous one
Training	Parallel and independent	Sequential and dependent

Basis	Bagging	Boosting
Data used	Random bootstrap samples (with replacement)	Same data, but misclassified points get higher weight
Main aim	Reduces variance (overfitting)	Reduces bias (and variance)
Combining	Simple voting / averaging	Weighted voting / weighted sum
Base learners	Strong, high-variance models (deep trees)	Weak models (shallow trees / stumps)
Examples	Random Forest, Bagged Trees	AdaBoost, Gradient Boosting, XGBoost

Why ensembles perform better than a single model

- **Error cancellation:** different models make different mistakes, and when their predictions are combined many of these errors cancel out, which reduces variance.
- **Bias reduction:** boosting converts many weak learners into one strong learner.
- **Robustness:** the result is less affected by noise, outliers and the specific training sample.
- **Better generalisation:** the combined model performs better on unseen data. This works best when the base models are reasonably accurate and different from each other.

Question 7 (5 marks)

Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick in SVM.

Answer:

Working principle

Support Vector Machine is a supervised algorithm used mainly for classification. For two classes of data, many different lines can separate them, but SVM selects the **optimal hyperplane** – the one that is farthest from the nearest data points of both classes. Such a hyperplane has the **maximum margin** and is less likely to make mistakes on new data, so it gives better generalisation. The hyperplane is written as $w \cdot x + b = 0$ and a new point is classified according to the sign of $w \cdot x + b$ (positive for one class, negative for the other).

Hyperplane: $w \cdot x + b = 0$ *Maximise margin* $2/||w|| \Leftrightarrow$ *minimise* $\frac{1}{2}||w||^2$ *subject to* $y_i(w \cdot x_i + b) \geq 1$

Margin and support vectors

- **Margin:** the distance between the two parallel lines $w \cdot x + b = +1$ and $w \cdot x + b = -1$ that pass through the closest points of each class. Its width is $2/||w||$, and SVM tries to maximise it.
- **Support vectors:** the training points that lie on the margin boundaries, i.e. the points closest to the hyperplane. Only these points decide the position and direction of the hyperplane; removing any other point does not change the model.

- **Soft margin:** when the classes overlap, a parameter C allows some misclassification. A large C gives a narrow margin with fewer errors, while a small C gives a wider margin that tolerates more errors.

Role of the kernel trick

Many real datasets cannot be separated by a straight line. In such cases, the data can be mapped to a higher-dimensional space where a linear separator exists. Doing this mapping explicitly is very expensive. The **kernel trick** solves this by using a kernel function K(xi, xj) that directly gives the dot product of the points in the higher-dimensional space without actually computing the transformation. In this way SVM can create non-linear decision boundaries efficiently. Common kernels are:

- Linear kernel: $K(x, z) = x \cdot z$
- Polynomial kernel: $K(x, z) = (x \cdot z + c)^d$
- RBF (Gaussian) kernel: $K(x, z) = \exp(-\gamma \|x - z\|^2)$

Question 8 (5 marks)

List and briefly explain any four-evaluation metrics used for classification problems (e.g., Accuracy, Precision, Recall, F1-Score, ROC-AUC). When is Accuracy not a good metric, and which metric would you prefer instead?

Answer:

The classification metrics are calculated from the **confusion matrix**, which has four values: **TP** (true positives), **TN** (true negatives), **FP** (false positives) and **FN** (false negatives).

Metric	Formula	Explanation
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Percentage of all predictions that are correct.
Precision	TP / (TP + FP)	Out of all cases predicted as positive, how many are actually positive. Important when false alarms are costly (e.g. spam filter).
Recall	TP / (TP + FN)	Out of all actual positive cases, how many were found by the model. Important when missing a positive is dangerous (e.g. cancer detection).
F1-Score	2 × (P × R) / (P + R)	Harmonic mean of precision and recall; gives one balanced score when both matter.
ROC-AUC	Area under the TPR vs FPR curve	Shows how well the model separates the classes at all thresholds. 0.5 means random guessing and 1 means perfect.

When is accuracy not a good metric?

Accuracy is misleading when the dataset is **imbalanced**, i.e. one class is much more common than the other. For example, if only 2 out of 100 transactions are fraud, a model that always predicts

"not fraud" will have 98% accuracy but it will never detect a single fraud, so it is useless. Accuracy also ignores the fact that different errors can have different costs.

In such cases we should prefer **Recall** (when missing a positive case is costly, like disease or fraud detection), **Precision** (when false alarms are costly) or the **F1-Score** / **ROC-AUC** (when we need a balance between both on imbalanced data). For highly imbalanced data, the F1-Score is generally the best single choice.

Question 9 (5 marks)

For regression problems, explain the following evaluation metrics: (a) Mean Squared Error (MSE) (b) Root Mean Squared Error (RMSE) (c) Mean Absolute Error (MAE) (d) R-squared (Coefficient of Determination). Which metric is more sensitive to outliers and why?

Answer:

Notation: y_i = actual value, $\hat{y}_i$ = predicted value, $\bar{y}$ = mean of the actual values, n = number of observations.

(a) Mean Squared Error (MSE)

$$MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$$

It is the average of the squares of the errors. Squaring makes every error positive and gives much more weight to large errors. Its unit is the square of the unit of the target variable, and a lower value means a better model.

(b) Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE} = \sqrt{[\ (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2\ ]}$$

It is the square root of MSE. Because of this it has the same unit as the target variable, so it is easier to interpret as the typical size of the prediction error.

(c) Mean Absolute Error (MAE)

$$MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|$$

It is the average of the absolute values of the errors. It also has the same unit as the target variable, treats every error in proportion to its size, and is easy to understand.

(d) R-squared (Coefficient of Determination)

$$R^2 = 1 - SSres\ /\ SStot = 1 - \Sigma(y_i - \hat{y}_i)^2\ /\ \Sigma(y_i - \bar{y})^2$$

It tells what proportion of the variation in the dependent variable is explained by the model. $R^2 = 1$ means a perfect fit, $R^2 = 0$ means the model is no better than simply predicting the mean, and it can even be negative for a very poor model.

Sensitivity to outliers

MSE (and therefore RMSE) is more sensitive to outliers. The reason is that the errors are squared before averaging, so a single very large error gets a very large weight and dominates the total. For example, an error of 10 contributes 100 to MSE but only 10 to MAE. MAE increases

only linearly with the error, so it is more robust. Hence, if the data contains many outliers MAE is preferred, and if large errors must be penalised heavily RMSE/MSE is preferred.

Question 10 (5 marks)

Discuss any four real-world applications of Supervised Learning. For each application, mention the type of supervised learning problem (classification or regression) and a suitable algorithm that can be used.

Answer:

Application	How it works	Problem type	Suitable algorithm
1. Email / SMS spam detection	The model is trained on messages labelled spam or not spam, using words, links and sender details as features, and then filters new messages.	Classification (binary)	Naive Bayes, Logistic Regression
2. House price prediction	Past sales data (area, location, number of rooms, age) is used to predict the selling price of a new property.	Regression	Linear Regression, Random Forest Regressor
3. Medical diagnosis	Patient reports, symptoms or scan images are used to predict whether a disease such as diabetes or a tumour is present.	Classification	Decision Tree, SVM, Random Forest
4. Credit card fraud detection	Past transactions labelled genuine or fraud are used to identify suspicious transactions in real time.	Classification	Random Forest, XGBoost

Other applications are sales and stock price forecasting (regression), weather and temperature prediction (regression), sentiment analysis of product reviews (classification) and face or handwritten digit recognition (classification).